Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21313

F-413

B.C.A. (Third Semester) Examination, 2023-24

(New Course)

Paper - BCA-303-N

DATA STRUCTURES USING 'C'

Time : Three Hours] ***[Maximum Marks : 70***

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Explain the various types of data structures.

   (b) Define the term recursion. Write a program in C to display factorial of a entered number using recursion.

2. (a) What are the major data structures used in the following areas: RDBMS, Network data model & Hierarchical data model?

   (b) What are the notations used in evaluation of Arithmetic Expressions using prefix and postfix forms?

3. (a) Describe working of selection sort method. Also sort given input list in ascending order using selection sort input list- 55, 25, 5. 15, 35.

   (b) Write an algorithm to count number of nodes in singly linked list.

4. (a) Construct a binary search tree for following elements: 30, 100, 90, 15, 2, 25, 36, 72, 78, 10 show each step of construction of BST.

   (b) Write algorithm for performing push and pop operations on stack.

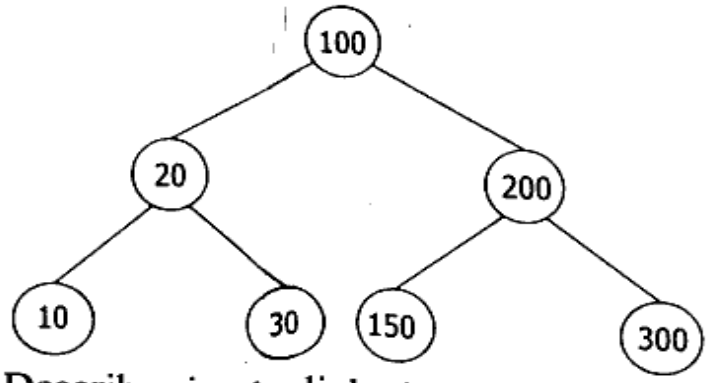
5. (a) Convert the following infix expression to its prefix form using stack A+B–C*D/E+F. Show diagrammatically each step of conversion.

   (b) Differentiate between Binary Search and Sequential Search.

6. (a) Find the position of element 29 using binary search method in an array 'A' given below : A = {11, 5, 21, 3, 29, 17, 2, 43}.

(b) Write an algorithm to delete a node from the beginning of a circular linked list.

7. (a) Traverse the following tree by the in order. Pre-order methods:



(b) Describe circular linked list with suitable diagram. Also state advantage of circular linked list over linear linked list.

8. (a) Describe working of bubble sort with example.

(b) Differentiate between general tree and binary tree.